



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Design and Simulation of a Full Adder

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1. Experiment Name

Design and Simulation of a Full Adder.

2. Objective

- To design and implement a Full Adder using Verilog HDL.
- To understand how the three inputs, **a**, **b**, and **cin**, produce the **sum** and **carry** outputs.
- To generate and observe the RTL schematic of the Full Adder in Xilinx ISE 14.7.
- To simulate the design using ISim and verify the output for all possible input combinations.
- To compare the simulation results with the expected Full Adder truth table.

3. Apparatus / Hardware & Software Requirements

- Xilinx ISE 14.7 — used to write and synthesize the Verilog HDL code and generate the RTL schematic.
- ISim Simulator — used to simulate the Full Adder and observe the output waveform.
- Verilog HDL — used to describe the logic and behavior of the Full Adder circuit.
- Oracle VirtualBox — used to run the Xilinx ISE virtual machine environment.

4. Circuit Diagram / Schematic

The Full Adder was described in Verilog HDL using three inputs, **a**, **b**, and **cin**, and two outputs, **sum** and **carry**. After synthesizing the design in Xilinx ISE 14.7, the RTL schematic was generated to observe how the design was represented at the logic level.

The logic of the Full Adder is based on the following expressions:

$$\text{Sum} = a \oplus b \oplus \text{cin}$$

$$\text{Carry} = (a \cdot b) + (b \cdot \text{cin}) + (\text{cin} \cdot a)$$

The **sum** output becomes high when an odd number of the three inputs are high. The **carry** output becomes high when at least two of the inputs are high.

Figure 1

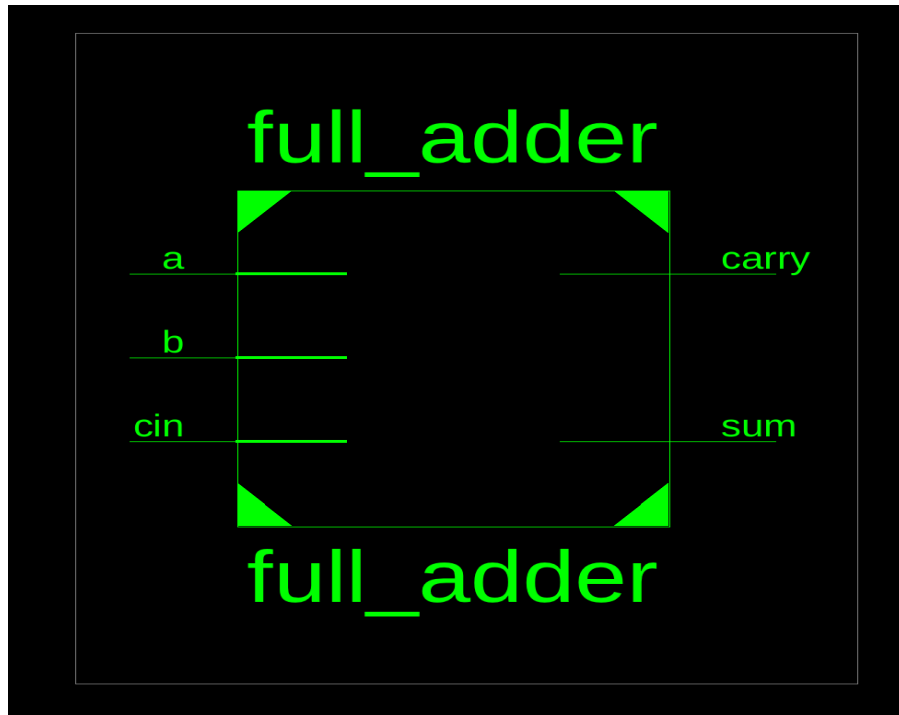


Figure 1: Top-Level RTL Schematic of the Full Adder

Figure 1 shows the top-level RTL view of the Full Adder generated in Xilinx ISE. The block contains three input ports, **a**, **b**, and **cin**, and two output ports, **sum** and **carry**.

Figure 2

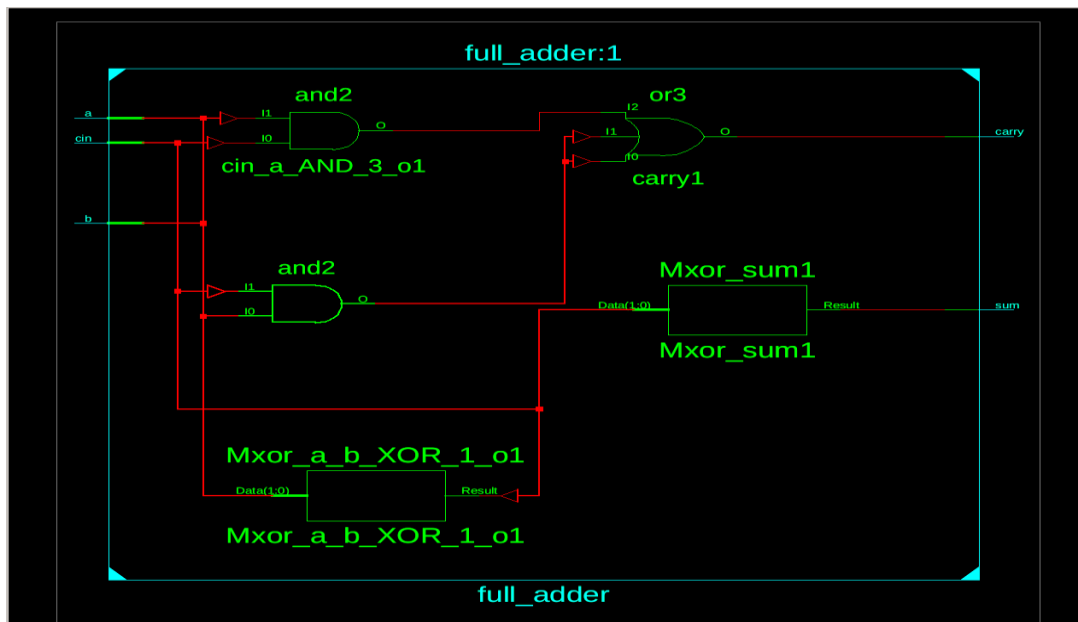


Figure 2: Expanded RTL Schematic of the Full Adder

Figure 2 shows the expanded RTL representation of the design. It gives a clearer view of the internal logic used to generate the **sum** and **carry** outputs from the three input signals. This schematic was obtained after synthesizing the Verilog design in Xilinx ISE 14.7.

6. Output Analysis / Observations

The Full Adder was simulated in ISim using all possible combinations of the three inputs **a**, **b**, and **cin**. Each input combination was applied for **100 ns**, allowing the corresponding **sum** and **carry** outputs to be observed clearly in the waveform.

The simulation showed that the **sum** output becomes high when an odd number of inputs are high. The **carry** output becomes high when at least two of the three inputs are high. The observed outputs matched the expected behavior of a Full Adder for all eight input combinations.

Simulation Results

a	b	cin	sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

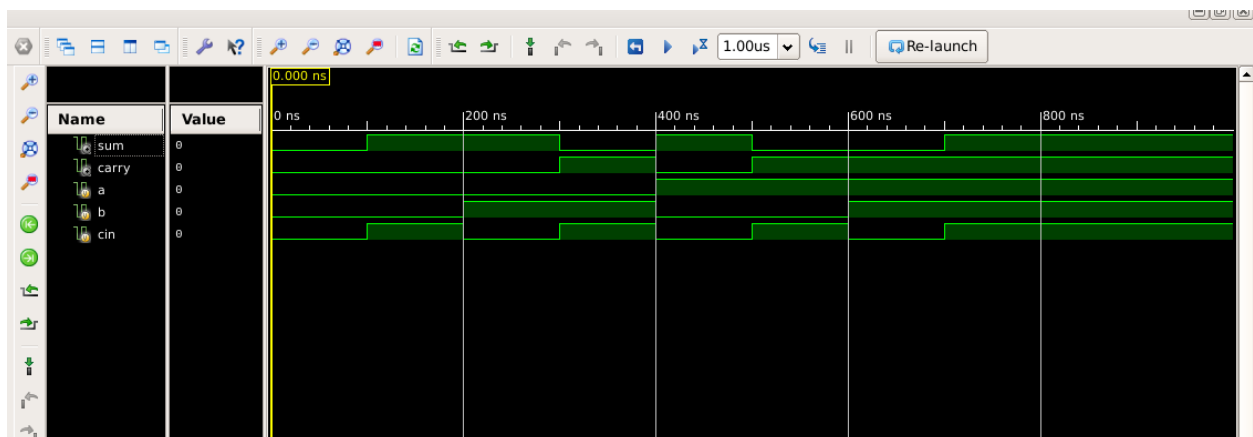


Figure 3: Simulation Waveform of the Full Adder

Figure 3 shows the ISim simulation waveform of the Full Adder. The inputs **a**, **b**, and **cin** were changed through all eight possible combinations, and the corresponding **sum** and **carry** outputs were observed. The waveform confirms that the implemented Verilog design produces the correct output for each input condition.

From the waveform, no unexpected output behavior was observed during the simulation. After the final test case, the last input and output values remained unchanged for the rest of the simulation time.

7. Result

The Full Adder was successfully designed and implemented using Verilog HDL in Xilinx ISE 14.7. The RTL schematic was generated after synthesis, and the design was simulated in ISim using all eight possible input combinations. The **sum** and **carry** outputs matched the expected Full Adder truth table in every case. Therefore, the objective of the experiment was achieved successfully.

8. Github Repository Commits

Commit: Add Full Adder design, RTL schematics, testbench, and simulation results

Commit Link: <https://github.com/naima-tabassum/CSE450-VLSI-Lab/commit/bbc7f5b0db48bffe3e289a8d65373dd2ae43f871>

9. Conclusion

This experiment gave practical experience in designing and verifying a Full Adder using Verilog HDL. The circuit was implemented in Xilinx ISE 14.7, and the generated RTL schematics helped show how the design was represented at the logic level. The design was then simulated in ISim using all possible input combinations, and the **sum** and **carry** outputs matched the expected Full Adder behavior. Overall, the experiment improved my understanding of how a digital logic circuit can be described in Verilog, synthesized, and verified through simulation.